



Human-in-the-Loop Analytical Leadership

A professional method for directing
AI-assisted analytical work.
Demonstrated through a synthetic financing case.

Andrew R. Goad

- 01 **AI**
accelerated the work.
- 02 **Human judgment**
directed the work.
- 03 **Evidence**
governed the work.
- 04 **Separate challenge**
strengthened the work.
- 05 **Teaching**
made it transferable.

THE CASE STUDY

Merchant Sales-Based Financing (MSBF)

Explore the strategy simulator and published project evidence

github.com/andrew-goad/merchant-sales-based-financing-strategy-simulator

Agree on direction. Delegate the work. Retain judgment.

This instruction followed context gathering and strategy refinement, before execution.

ORIGINAL HUMAN INSTRUCTION / AUG. 28, 12:02 PM / EXCERPT

One more thing before we begin. You have ingested the entire GitHub (including project history/lineage), data, Power BI, Campaign work, Module 3 work, credit decisioning strategy simulator code. You have all the keys to the kingdom, so to speak to make this successful. I agree with the direction you've outlined. I want to do this work in as few chats prompt interactions as possible and in as few days as possible. Issues I've noticed in the past - We do not want to waste time/resources on a prompt that is too large and doesn't complete. We also do not want to spend time doing too little in each response and taking 100 chats to do something we could have done in 5. We want to maximize what we accomplish with each prompt. I authorize you to move ahead in leaps and only stop when you need a key decision or task from me. If it's something we've already agreed on in the strategy, do not raise that to the level of a decision. I agree with your recommendations now and the direction we've set now. We do not have to establish the full roadmap now, but please keep that as a governing principle and know that we will not go back to audit until just before republishing.

CONTEXT SUPPLIED

EXECUTION DELEGATED

JUDGMENT RETAINED

Follow the decisions that changed the work.

Read the headlines and teaching notes for the story. Inspect the original interactions for the evidence.

PAGES	THE METHOD IN PRACTICE	THE SKILL DEMONSTRATED
CYCLE 1 / Improve the decision process		
04-08	A Follow the missing explanation	Trace the evidence and find the cause
09-10	B Challenge the accepted rule	Challenge rules and adapt prior work
11-17	C Coordinate, execute and test	Coordinate AI work and define the test
18-19	D Obtain a bounded acceptance	Seek review and respect its limits
CYCLE 2 / Reopen the monitoring question		
20-25	E Keep questioning after acceptance	Question measures and test an explanation
26-29	F Verify the correction	Verify the change and preserve the record
RESULTS / Explain and teach the method		
30-35	G Explain what changed	Explain the result and teach the method

HUMAN DIRECTION Set direction. Retain decisions.	REPORTING / PRODUCT Expose behavior. Define reporting needs.	REMEDIATION / BUILD Investigate, develop and validate.	SEPARATE AUDIT Challenge and bound acceptance.
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I coordinated the separate conversations and retained responsibility for business decisions.
The chats did not communicate autonomously.

Pages 7-8 pair later Merchant 003 outcomes with the original question. Event dates remain beside the evidence; chronology details are in the source guide.

What evidence was there - and why was it blocked?

Merchant 003: Baseline provides a detailed write-up. Recession Energy names a block without explaining the evidence behind it.

ORIGINAL HUMAN PROMPT / EXCERPT

Here is Merchant 003. You see detailed narrative for Baseline, but the same required underwriting was blocked... for recession energy. I need more than that. What evidence was there, and why did it get blocked before a complete operating assessment could be produced. We need that why in our return. It's not sufficient to say, "Required source or verification evidence was insufficient to support a complete operating assessment." as you suggest.

ORIGINAL REPLY / SAME CAPTURE / EXCERPT

Yes. Merchant 000003 exposes the next issue: we are still narrating the status of the pipeline rather than the evidence chain that caused that status.

For 000003 under Recession Energy, we actually know considerably more than the page is saying. The selected row has high source confidence (98.6%), clear verification, fraud Tier 1, monitored processor continuity, complete feature-completeness status, \$28.6K average available balance, no negative-balance days, and no NSF's. What disappears under stress is the merchant's 30-day eligible-sales measure and annualized eligible sales. Once `avg_daily_eligible_sales_30d` is unavailable, M1.10 cannot estimate daily operating cash flow; the governed capacity engine explicitly sets capacity evidence to `BLOCKED` when that sales measure is null. Coverage and residual cash flow therefore cannot be calculated, and the missing capacity evidence cascades into resilience/risk/loss/economics being unusable.  Pasted text(20260827-165737)

AUG. 27 / 11:54 PM

THE HUMAN INTERVENTION

Baseline describes the merchant's operating facts. Recession Energy says evidence was blocked. The human question asks what was available, what was missing and why that stopped the assessment.

PRACTICE TO REUSE

A blocked assessment still needs a reason the reader can trace to source evidence.

Baseline explained the assessment. The stress view did not.

One synthetic applicant under two economic scenarios. These are alternatives, not successive events.

BASELINE / ORIGINAL WRITE-UP

① **Supporting detail is visible.**

Merchant operating evidence

The merchant showed a volatile operating pattern, averaging \$4.0K in eligible sales per day, with no recent NSF events and no negative-balance days. Average available balance was \$99.3K. Processor continuity required monitoring. The evidence was high-confidence and complete, with clear verification.

Structure assessment narrative

At the requested structure, the governed capacity assessment was unaffordable: modeled operating cash flow covered total daily obligations 0.73x, producing a \$360 per-day shortfall, with a post-financing liquidity buffer of \$88.5K. Integrated risk was moderate (Tier 2; score 21.1). Comparative expected loss was 6.37%. Risk-adjusted contribution was \$2,168 (21.5% annualized), above the governed hurdle.

P03 Decision Readout

The requested structure exceeded sustainable cash-flow capacity; no supportable alternative structure was authorized.

Sales, balances and cash-flow calculations give the reader evidence behind the assessment.

RECESSION ENERGY / ORIGINAL WRITE-UP

② **The explanation stops at “blocked.”**

Merchant operating evidence

Required underwriting evidence was blocked before a complete operating assessment could be produced.

Structure assessment narrative

Required underwriting evidence was blocked before the governed capacity, risk, comparative-loss, and economic assessment could be completed.

P03 Decision Readout

One or more governed eligibility or structure gates failed; no supportable structure was authorized.

Which evidence was unavailable? Why was it unavailable? Why did that stop the assessment?

A blocked assessment still needs an evidence-based explanation.

Move an unresolved explanation to the right workstream.

The human prompt asks the reporting conversation to prepare a handoff to the original build chat.

HUMAN PROMPT / OPENING + QUOTED AI CONTEXT

please draft a detailed prompt to deliver back to my original build chat, regarding this: "There is a model-design issue we should acknowledge There are really two levels of "why":

1. Why did capacity become blocked?

We can answer this from FactApplicationScenario : avg_daily_eligible_sales_30d is blank, and the capacity engine explicitly blocks when that field is unavailable.

2. Why did the 30-day eligible-sales feature itself become unavailable?

FactApplicationScenario does not contain enough diagnostic lineage to answer that safely.

THE CLOSING QUESTION / VERIFIED TRANSCRIPT

“It seems like something we should investigate as a potential Module 1/2 issue. Would you agree?”

AI RESPONSE / VERIFIED TRANSCRIPT

“Yes. I agree this warrants a bounded cross-layer investigation as a potential Module 1/Module 2 issue, rather than another Power BI wording adjustment.”

WHAT THE INSTRUCTION CHANGED

The unresolved question moved from report wording to source investigation.

The prompt quotes earlier AI analysis. The human contribution shown here is the request to escalate and obtain a bounded handoff.

A \$42,500 alternative emerges. Review still governs the decision.

An alternative offer appears in the later Baseline view. The original Recession Energy question is answered on page 8.

ORIGINAL POWER BI DETAIL / EDITORIAL CUES 1-2

1

Merchant request

Applied	Requested	Use	Channel	Remittance	Payback multiple	Payoff days	Repayment
7/23/2026	\$61,300.00	Working capital	Digital direct	13.67%	1.15	60	\$70,644.20

REVIEW

Counteroffer review

Proposed structure

Amount	Remittance	Payback multiple	Horizon days	Repayment	Funding date
\$42,500.00	13.67%	1.15	60	\$48,978.45	

Merchant operating evidence

This is a new lower-middle-market healthcare-services merchant sourced through the digital-direct channel. Recent decision-time evidence showed approximately \$1,278,686 in annualized eligible sales. Average available liquidity was approximately \$99,341. No NSF events or negative-balance days were observed in the governed 30-day window.

Risk & economics assessment

The merchant requested \$61,300 at a 13.67% remittance rate, 1.15x payback and a 60-day horizon, for \$70,644 in total repayment. The original terms required adjustment, and the strategy evaluated 11 governed alternatives before identifying a review-eligible structure. The selected alternative changed the amount from \$61,300 to \$42,500. The review-eligible terms provide 1.00x payment coverage, retain approximately \$99,376 in modeled post-financing liquidity, and produce \$2,543 in risk-adjusted contribution at a 4.1% comparative expected-loss rate. The structure is not yet authorized because the proposed structure sits near the approved capacity boundary.

Decision

2

Counteroffer review required. A potentially supportable \$42,500 alternative was identified. The proposed structure sits near the approved capacity boundary, so specialist review is required before terms can be authorized. No funding has occurred.

1 AN ALTERNATIVE

The original request was \$61,300. The later system identifies a potentially supportable \$42,500 offer.

2 REVIEW REQUIRED

The offer still needs specialist review. It has not been authorized, and no funding has occurred.

Original Baseline decision-column crop; editorial outlines and notes. Later corrected result paired with the earlier question.

CONTENTS

07

The stress view now explains the block.

Later Recession Energy result: the write-up identifies the missing sales evidence and why the assessment cannot proceed.

ORIGINAL POWER BI DETAIL / EDITORIAL CUES 1-2

DECLINED

Evidence decline

Merchant request

Applied	Requested	Use	Channel	Remittance	Payback multiple	Payoff days	Repayment
7/23/2026	\$61,300.00	Working capital	Digital direct	13.67%	1.15	60	\$70,644.20

Proposed structure

Amount	Remittance	Payback multiple	Horizon days	Repayment	Funding date
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Merchant operating evidence

1

This is a new lower-middle-market healthcare-services merchant sourced through the digital-direct channel. The merchant profile was available, but the governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days. Because that gap prevented a reliable capacity and risk assessment, the platform stopped rather than treating missing evidence as weak performance.

Risk & economics assessment

The merchant requested \$61,300 at a 13.67% remittance rate, 1.15x payback and a 60-day horizon, for \$70,644 in total repayment. A reliable structure assessment could not be completed because the governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days. This is an evidence-based stop, not a conclusion that every possible structure would fail.

Decision

2

Declined because required evidence was unavailable. The governed 30-day sales window contained 28 usable days versus 30 required, including 2 disconnected processor-outage days, so the platform could not establish a sufficiently reliable view of the merchant's operating position. No structure or funding decision was inferred from incomplete data.

1 THE SPECIFIC GAP

Only 28 of the required 30 sales-history days are usable. Two days were lost to processor outages.

2 THE EXPLAINED STOP

The missing sales history blocks the dependent assessment. That explains the stop; it does not prove every offer would fail.

Question the rule behind a correct implementation.

The system followed its accepted rules. The next question was whether those rules led to sensible decisions.

PRIOR AI FINDING / MERCHANT 003

“Yes under the currently accepted routing rules.”

HUMAN INSTRUCTION / AUG. 28, 10:56 AM / VERIFIED TRANSCRIPT

“I agree with your conclusions and analysis. I also authorize a change to the structure of this chat. We will go to audit, but I prefer to proceed with remediation and having audit review post remediation. Before we remediate, I'd like to discuss your answer to this question: Is Merchant 000003's final outcome correct?

Yes under the currently accepted routing rules. Do you agree with the currently accepted routing rules? Notably, I was surprised that so few applications were accepted in our 750 initial population. Also, I note a high review population.”

The next retained instruction supplies prior counteroffer code and defines the correction. The low acceptance and high review population prompted a question about the decision path.

Bring relevant prior work into the correction.

Could a different permitted offer work? The instruction brings prior counteroffer code into the investigation.

ORIGINAL HUMAN INSTRUCTION

`</>` module2_credit_policy_strategy_decis...
File

`</>` module1_synthetic_application_risk_e...
File

We built significant counteroffer logic in my credit decisioning simulator. I was surprised at the lack of sophistication in the MSBF counteroffer logic. I provided my credit decisioning simulator to the initial build as an example. Here is my credit decisioning simulator logic again to ingest. After ingesting, I propose that we change the scope of this chat. This chat will be dedicated to correcting the module 1 /2 strategy to 1) find an affordable counteroffer before declaring that no supportable structure exists (using similar gate / threshold logic as in the credit decisioning simulator) ; 2) correcting the issue our Power BI chat called out ; 3) any other strategy/defects. The goal then will be to publish a new GitHub Module 1 / 2 version and release and also feed Power BI everything it needs to then continue (after those above items are complete).

AUG. 28 / 11:08 AM

THE ASSIGNMENT

Find an affordable alternative before declaring that no supportable structure exists. Correct the evidence issue and return what Power BI needs to continue.

PRACTICE TO REUSE

Supply the design reference and define the problem it should help solve.

Design the handoff before the rerun.

Plan how the work will run and what evidence it will leave, before asking AI to execute it.

ORIGINAL HUMAN INSTRUCTION / ATTACHED CONTEXT



MSBF_CAMPAIGN_SCALE_TO_M1_M2_...
Document



MSBF_CAMPAIGN_SCALE_TO_M1_M2_...
File

Let's think about evidence we want to capture in this re-run and how we do the re-run. We had a separate chat called MSBF - Campaign Scale ... that has been working towards a smoother way to perform a full re-run and what evidence is important. Here is the latest from that chat (although notably, audit drove it sort of madly into a spiral around overaggressive bootstrapping/keys...versus getting the job done). Is anything usable from that pursuit? I need to find a better way than running 230 odd programs back to back to ... We will follow my no cold handoffs approach, but in this instance, I will be heavily involved in filling audit's shoes upfront and we'll pass everything we did to audit afterwards to approve before publishing. The most important thing is accuracy and speed and maintaining the high professionalism you see in my credit decisioning and msbf work to date. This should be exceptional.

AI RESPONSE / VERIFIED TRANSCRIPT

“The rerun should be one command, not 230 manual executions”

The proposed approach combines one controlled run with ordered execution and traceable evidence underneath.

PRACTICE TO REUSE

Carry forward useful controls while reducing avoidable execution burden.

Bring reporting requirements into the rebuild.

Ask the reporting workstream what the rebuilt system must explain before implementation begins.

ORIGINAL HUMAN INSTRUCTION

I have started a MSBF - Module 1/2 Strategy Correction, Remediation, & Republishing chat to correct the defect you found and also to correct an overly conservative strategy leading to low acceptance/low level of counteroffers. The MSBF lacked the counteroffer sophistication I had implemented in my credit decisioning strategy simulator github repository. Please provide a summary of where we'd like our power BI model to progress towards down the road for this new chat to consider while making changes. I have already provided the Power BI project folder to the new chat showing our current Power BI model.

AI RESPONSE / OPENING EXCERPT

Yes. The new correction/remediation chat should treat the current Power BI carrier not merely as something that must continue to load after the Module 1/2 changes, but as an important downstream consumer whose future analytical needs should influence what explanation, strategy, offer, economics, and lifecycle evidence the rebuilt source contracts preserve.

AUG. 28 / 11:37 AM

THE COORDINATION

A remediation chat is underway. The prior simulator and Power BI project are already supplied. The instruction asks for future reporting requirements before the source is rebuilt.

PRACTICE TO REUSE

Carry the product question and evidence requirements into the source assignment.

Carry the reporting feedback back into the source strategy.

The reporting chat defines what users need to understand. The owner carries those requirements back to the build chat.

REPORTING RESPONSE / ORIGINAL REQUIREMENT EXCERPT

Power BI should ultimately be able to show:

Requested structure → why it failed or became marginal → feasible structure search → candidate alternatives → governing constraints → selected counteroffer → economics/risk improvement → review/authorization outcome

HUMAN RETURN HANDOFF / ORIGINAL PROMPT + ATTACHMENT

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Document

Feedback from our Power BI chat before we embark:

REMEDATION RESPONSE / ORIGINAL OPENING EXCERPT

This Power BI feedback is highly usable, and I would incorporate most of it before we write the successor strategy. It does not materially change the direction we were headed; it sharpens the evidence contract so that we do not repeat the same mistake—building correct-enough strategy logic but leaving downstream consumers unable to explain it.

Turn delegation into stopping conditions.

The response to the page 2 mandate specifies when work can continue and when a human decision is needed.

AI RESPONSE TO THE MANDATE / ORIGINAL EXCERPT

Understood. I'll operate this workstream on that basis.

From here forward, I'll optimize for **maximum useful progress per interaction without creating oversized, failure-prone actions**. I'll make routine implementation choices myself when they follow from the strategy we have already agreed on, and I will not keep stopping to reconfirm settled principles such as counteroffer-before-decline, exact lineage preservation, integrated evidence capture, Power BI explainability, predecessor-vs-successor reconciliation, clean rerun architecture, or post-remediation audit.

I'll stop only when one of four things is true: a genuinely new business-policy choice would materially change the strategy; I need an external action or artifact from you; available evidence leaves two materially different defensible paths; or continuing would cross an authority boundary you have not granted.

Practically, that means I will structure the work into **large but bounded leaps**. A typical response should complete a coherent development unit—for example, successor strategy design plus population counterfactual analysis, or implementation plus validation—not merely propose the next tiny step. Conversely, I will split an action before it becomes so large that execution reliability or evidence quality degrades.

AI RESPONSE / BUILD VALIDATION AND SEPARATE AUDIT

And the audit operating model is now clear: we **build, challenge, rerun, reconcile, validate, package, and mature the successor ourselves**; Independent Audit re-enters immediately before republication to review the completed candidate rather than co-design it.

Build-stage challenge continues; the separate audit reviews a mature candidate.

WHAT THE RESPONSE DOES

Proceed with settled choices.
Return for a new policy decision, required external action, materially different paths or ungranted authority.

PRACTICE TO REUSE

Delegate substantial work.
Keep consequential judgment available as execution reveals new evidence.

Define the decision rules. Test them on the same applications.

The assignment asks how each request should be handled, which alternatives should be tested and what changes across 750 applications.

ORIGINAL HUMAN INSTRUCTION

Define the corrected decision state machine and counteroffer methodology, test it counterfactually against the accepted 750-application population, identify any additional strategy defects exposed by that analysis, and return with the proposed successor strategy and quantified impacts—stopping only if that work surfaces a genuine policy choice that requires you.

AUG. 28 / 12:04 PM

THE TEST REQUEST

Define the sequence of decisions and the search for alternative offers. Test the same 750 applications. Quantify changes and return genuine policy choices.

ORIGINAL AI FINDING / SAME CAPTURE

Corrected M1/M2 successor strategy

Executive determination

The accepted predecessor architecture contains a material strategy defect, not merely conservative calibration:

It evaluates capacity and affordability at the merchant's requested structure, then allows that requested-structure result to determine the merchant's final eligibility route before conducting a complete search for a supportable alternative.

That sequencing confuses two different conclusions:

- 1. The requested structure is not supportable.
- 2. No supportable structure exists for the merchant.

The first conclusion does not establish the second.

PRACTICE TO REUSE

Specify the question, comparison and decision boundary before implementation.

407 additional searches. Approval still requires authority.

A failed request does not prove that no permitted alternative exists. The early test identifies cases that need that search.

ORIGINAL RESPONSE / COMPLETE CAPTURE

Exact state-machine counterfactual

Under the corrected state machine:

Population	Current treatment	Correct successor treatment
101 capacity PASS	Offer or review	Retain requested candidate; evaluate residual noncapacity gates
82 capacity REVIEW	Manual review	Conduct structure search; retain review only when a genuine human-judgment gate remains
407 capacity FAIL	Immediate policy decline	Conduct complete counteroffer search
43 evidence-blocked	Insufficient-evidence decline	Preserve insufficient-evidence treatment; do not search
117 hard-stop	Policy decline	Preserve hard-stop decline; do not search

Therefore:

Mandatory Baseline structure-search population

= 407 FAIL + 82 REVIEW

= 489 applications

= 65.2% of the initial population

The corrected architecture creates **407 additional searches** that the predecessor never conducted.

It does **not** mean 407 additional approvals. It means 407 applications will no longer be declared unsupported without first proving that no feasible alternative exists.

The total decisionable, non-hard-stop population is:

101 PASS + 82 REVIEW + 407 FAIL

= 590 applications

= 78.7% of the population

The remaining 160 applications—21.3%—are blocked or governed hard stops and should not be pushed through counteroffer search.

407
additional searches

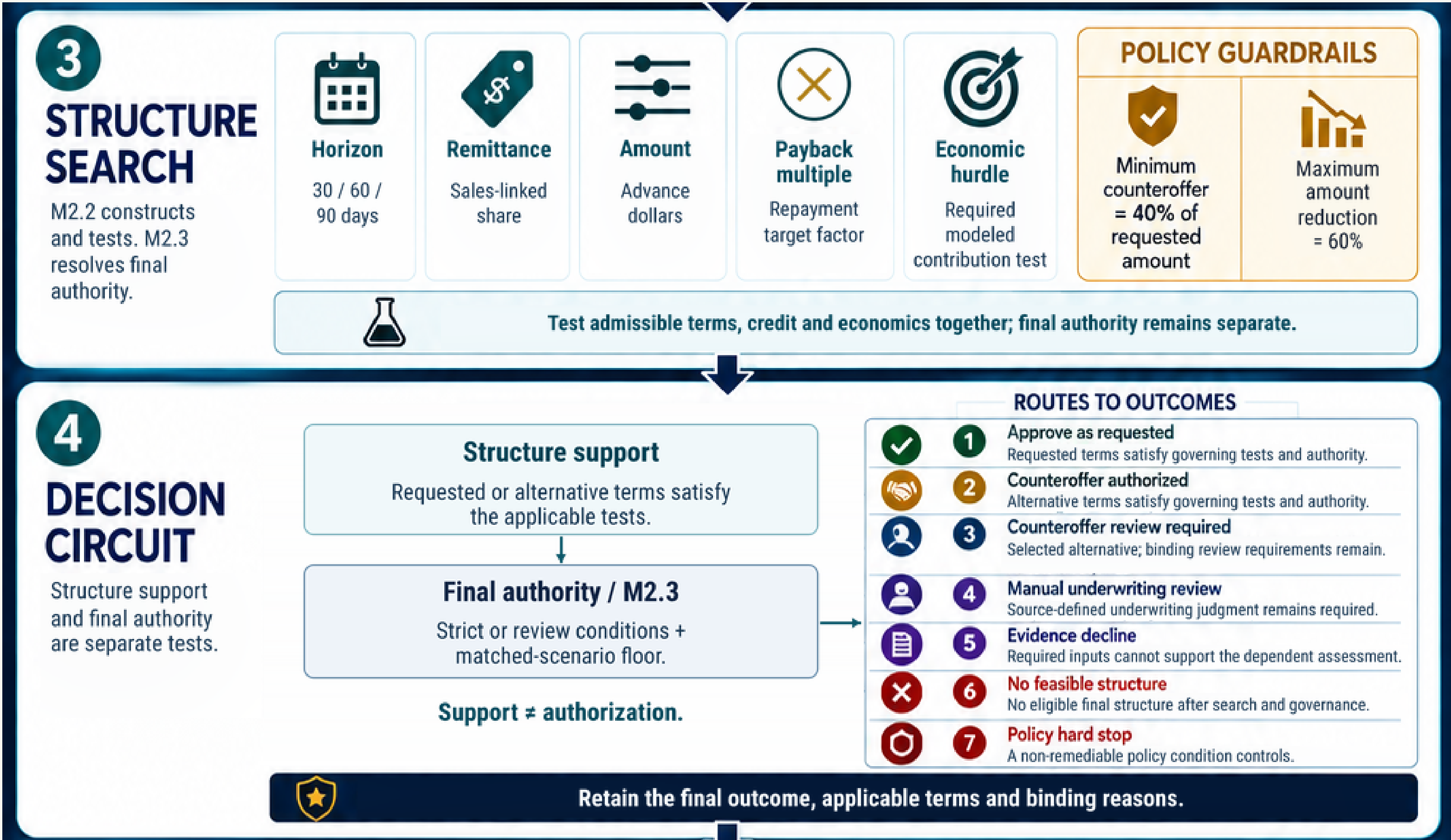
The initial diagnosis calls for 489 searches: 407 FAIL cases plus 82 REVIEW cases.

Search for an alternative. Check whether it can work. Then apply the separate approval rules.

An offer can fit the numbers and still require approval.

The revised design searches for permitted alternatives while preserving the rules about who may authorize them.

ORIGINAL GRAPHIC / SELECTED PANELS



Read the two panels as two questions: can an offer work within the rules, and is it authorized to proceed?

Give the completed candidate a separate challenge.

The first separate review (R23) accepts the revised build with observations. Acceptance establishes a reviewed milestone.

ORIGINAL R23 AUDIT FINDING / EXCERPT

R23 bounded successor audit completed

Decision ID: IA-D-R14-M12-001

Governing operating model: IA-GOV-20260822-001

Overall disposition: PASS_WITH_OBSERVATIONS

Recalibration determination

Immediate gate

Accept or reject the exact Module 1/2 R1 successor and determine whether a separate controlled republication action may proceed.

Material risks tested

Incorrect R17 or R19 results
Accepted-predecessor mutation
Repository/source divergence
Power BI strategy-method reconstruction or P01-P04 regression
Stale public authority state
Hosted-release packaging uncertainty

Further technical-remediation cycle required

NO

Controlled republication preferable to additional static assurance

YES

Audit burden proportionality

SATISFIED

No material strategy, policy, source, repository, or Power BI contradiction remains. The five observations are bounded release-finalization, historical-evidence, post-publication, and downstream-authority matters. They do not justify another technical remediation cycle.

THE EARLIER HUMAN INSTRUCTION

“We will go to audit, but I prefer to proceed with remediation and having audit review post remediation.”

FIRST CYCLE / ACCEPTED WITH OBSERVATIONS

The revised decision process reaches a defined acceptance milestone. Later dashboard evidence will raise a new question about monitoring.

Check the facts from source to report. Keep the limits visible.

The review asks whether the underlying records, simulated account history, saved build and dashboard agree.

ORIGINAL PHYSICAL RE-PERFORMANCE

1. R17 / Wave A

Disposition: PASS

The accepted R17 identity is:

eee512f19d38e189f811abc788b33ae8495ed1161be58c3861d5dc81f86b13be



Independent re-performance established:

Validation controls	115 / 115 PASS
M2.5 monitoring rows	15,688
Account paths	138
Monitoring rows per path	128
M2.6 through M2.18 rows	138 each
BASELINE paths	183
RECESSION_ENERGY paths	27
Transformed-source parity	12 / 12 exact
Source database modified	FALSE
Accepted authority changed	FALSE



The retry-settlement correction is physically represented. The known returned-payment path reconciles through M2.8 and M2.9 with zero downstream reconciliation and exposure variance.

REVIEW SCOPE / EDITORIAL SUMMARY

Underlying account and servicing records.
Simulated history and validation checks.
The saved build and its reconstruction.
Dashboard counts and reporting.

THE ACCEPTANCE BOUNDARY

The review supports a specific technical acceptance and next step. It does not establish real-world lending performance or unrestricted publication authority.

WHAT ACCEPTANCE ESTABLISHED

Agreement from source records through reporting, within the tested scope.
Later evidence can still raise a new business question.

An accepted build still has to make business sense.

A separate project review had accepted the build with observations. Later dashboard evidence raised a new monitoring question.

HUMAN FOLLOW-UP / INCLUDES QUOTED AI ANALYSIS

let's not leave this. let's dive in to understand it: One observation, not a defect

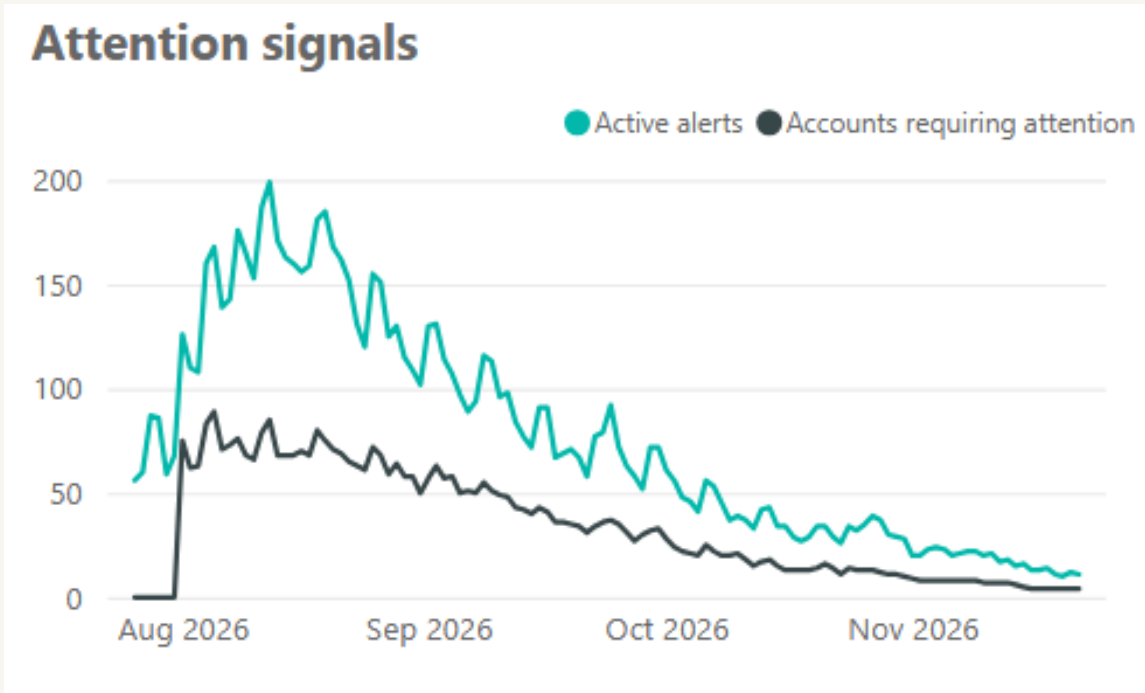
The P02 Attention signals chart has much larger absolute counts than the predecessor page. That initially looks dramatic, but the legend now explicitly says Active alerts and Accounts requiring attention, and the successor population is also much larger. The evidence shows those signals subsequently resolve toward zero. I would leave it exactly as built rather than normalize or suppress the behavior. first, specifically what are you comparing?

PRIOR AI RESPONSE / SAME CAPTURE

One observation, not a defect

The P02 Attention signals chart has much larger absolute counts than the predecessor page. That initially looks dramatic, but the legend now explicitly says Active alerts and Accounts requiring attention, and the successor population is also much larger. The evidence shows those signals subsequently resolve toward zero. I would leave it exactly as built rather than normalize or suppress the behavior.

THE PATTERN UNDER REVIEW / BEFORE



NEW QUESTION / AFTER ACCEPTANCE

Why do so many accounts appear to need attention? First, establish what the explanation is comparing.

Acceptance is a milestone. Keep judgment active as new behavior becomes visible.

Resolve the metric. Keep the business question open.

One account can raise several alerts. Clarifying that count still leaves the question: does the account behavior make sense?

ORIGINAL HUMAN FOLLOW-UP

I see, the old P02 was prior to the Remediation Corrections, so this would make sense. The Remediation Corrections made significant changes. The more logical question is do the current attention signals make sense for a population of this size and this type of product?

ORIGINAL AI EXPLANATION / EARLIER IN THE CAPTURE

That can be perfectly valid if "Active alerts" counts alert flags rather than accounts. One account can simultaneously carry multiple alert conditions. In fact, we named the R1 measure Active Alert Flags As of , which suggests exactly that.

But that means the chart is currently juxtaposing:

number of alert flags

with

number of account paths requiring attention

on the same count axis.

SEPT. 1 / CONTINUED REVIEW

COUNT CLARIFIED / CONCERN OPEN

Alert counts and account counts are different: one account can raise several alerts. The human follow-up still asks whether the current pattern makes sense for these accounts.

PRACTICE TO REUSE

Clarify the measure, then return to the business concern. Next, check the source records.

The early warning pattern is present in the underlying data.

A check of the source records reproduces the day-14 pattern. Explaining how the monitoring policy produced it still requires investigation.

ORIGINAL SOURCE LOG / REPRODUCTION EXCERPT

Scope and reproduction

Before changing any assumption, I independently reproduced the physical R17 result from the accepted R4 terms and accepted M1.6 histories:

Eligible-sales cent difference	0
Actual-remittance cent difference	0
Cumulative-remittance cent difference	0
Receivable cent difference	0
Cumulative-shortfall cent difference	0
Pace difference at 1e-8	0
Monitoring-status mismatches	0
Stress-floor mismatches	0
M2.6 endpoint-outcome mismatches	0

The accepted M2.5 method activates WATCH on day 7 and the higher-severity statuses on day 14, using 90%/75%/50% cumulative-pace thresholds, 7/14-day no-remittance thresholds, and a 10-day zero-sales threshold. M2.5 is expressly monitoring-only and does not authorize servicing actions.

BASELINE / MONITORING DAY 14

41 / 103

funded paths in
higher-severity statuses

- 27 underperforming
- 3 severe shortfall
- 11 dormant / no remittance

The early pattern remained after checking accounts rather than multiple alert flags. The source records confirmed the classifications; they did not establish that the policy was well calibrated. Further checks came before the timing hypothesis.

Pause presentation work while source behavior is unresolved.

Repeated checks have not resolved the concern. The owner pauses dashboard work while the build workstream investigates.

ORIGINAL HUMAN INSTRUCTION

pausing this power bi work until remediation/correction chat examines further. it seems a strategy update needs to be made.

ORIGINAL AI RESPONSE / SAME CAPTURE

Agreed. Pause the Power BI refinement here. We have enough evidence that continuing P02—and especially proceeding into P03/P04 lifecycle interpretation—could cause us to build presentation logic around monitoring semantics that may change upstream.

The broader QA examined 61 valid replay windows, eight expected paths and three persistence variants. An interim no-policy-change recommendation did not settle the question.

SEPT. 1 / FURTHER QA

THE STOPPING RULE

The prompt pauses Power BI until the remediation conversation examines the behavior. The response recognizes that presentation logic could otherwise be built around semantics that change upstream.

PRACTICE TO REUSE

Pause a downstream deliverable when unresolved source behavior could change its meaning.

Question whether the clock fits the funded account.

After diagnostic work and repeated checks, the human hypothesis asks whether newly funded accounts are being judged too early.

ORIGINAL HUMAN HYPOTHESIS

Tue, Sep 1 at 8:54 PM

is it possible that we are starting the 14 day timer too early after funding. perhaps the first month issue is related to funding timing and really a false positive. i do not want to delay this high priority issue. we need a policy/database solution now.

ORIGINAL AI SOURCE FINDING / SAME CAPTURE

So yes: the current pace-based deterioration clock becomes severe very early in a newly funded account's life.

More importantly, the current method compares two different concepts. Actual remittance is correctly sales-linked—`eligible POS sales × remittance rate`. `MSBF_R1_SUCCESSOR_REPOSITORY_CA...` But cumulative expected remittance is a fixed straight-line benchmark using `implied_daily_collection_amount × monitoring_day_index`. `MSBF_R1_SUCCESSOR_REPOSITORY_CA...` Then WATCH starts on day 7 and UNDERPERFORMING/SEVERE pace logic starts on day 14. `MSBF_R1_SUCCESSOR_REPOSITORY_CA...`

That combination is the issue I would correct now—not funding dates themselves.

SEPT. 1 / 8:54 PM

THE CAUSAL QUESTION

Does the system allow enough time after funding, and is the repayment benchmark appropriate? The response identifies a broader problem in how pace is judged.

PRACTICE TO REUSE

Offer a mechanism the source logic and retained evidence can test.

Observe from day one. Judge pace with maturity.

Continue collecting evidence daily. Adjust the repayment benchmark, the first eligible day and how long a pace breach must persist.

ORIGINAL POLICY PROPOSAL / COMPLETE CAPTURE

Recommended policy/database correction

I recommend we leave M2.4 funding, activation, and first-remittance dates alone and version M2.5 to a new monitoring policy with this profile:

Policy element	Current	Recommended successor
Daily evidence collection	Day 1	Day 1 — unchanged
Payoff-pace benchmark	<code>implied_daily_collection_amt</code>	<code>total repayment ÷ selected horizon</code>
Pace WATCH eligibility	Day 7	Day 21
Pace UNDERPERFORMING eligibility	Day 14	Day 30
Pace SEVERE eligibility	Day 14	Day 30
Pace persistence	Immediate	3 consecutive eligible days
7/14-day no-remittance logic	Existing	Preserve
10-day zero-sales logic	Existing	Preserve
Liquidity stress	Existing	Preserve
Horizon overrun	Existing	Preserve
Raw daily signals	Persisted	Preserve from day 1
Operational alert counting	Overlapping raw conditions	Separate non-overlapping alert families/account state

The important distinction is: **we do not stop monitoring for 30 days**. We observe everything from day 1. We simply stop treating volatile early payoff pace as mature performance evidence before the account has seasoned.

Hard exceptions remain live. A merchant with 14 days of no remittance, a severe zero-sales streak, liquidity failure, or horizon overrun can still be identified early.

21 First eligible day for WATCH

30 First eligible day for adverse pace

3 Consecutive eligible days in breach

Daily evidence and other hard triggers remain. Earliest pace-only qualification: day 23 for WATCH; day 32 for adverse pace.

Request a bounded challenge. Receive a recorded decision.

The second focused review, identified as R26, checks the monitoring correction, its outputs and the permitted next step.

ORIGINAL AUDIT REQUEST / EXCERPT

COPY/PASTE PROMPT — BOUNDED R26 MONITORING-CALIBRATION SUCCESSOR AUDIT

Perform a focused Independent Audit of:

MSBF_R26_FOCUSED_INDEPENDENT_AUDIT_SUBMISSION_20260902.zip

Before deciding:

1. verify the outer manifest/checksum/receipt;

2. ingest 00_START_HERE.md, all docs/, prior_authority/, evidence/, and required inputs/;

3. treat the prior R23 decision as controlling predecessor acceptance authority;

4. independently verify the physical R26 R2 failure result and final resume PASS result rather than reconstructing them from this prompt;

5. preserve accepted/public predecessor authority unless this focused audit explicitly accepts R26 as the next technical successor baseline.

Decision requested: accept/reject/hold R26 M2.5–M2.12 as a technical successor baseline, including the owner-approved monitoring policy, the retained-final resume method, corrected M2.11 semantic identity boundary, 52,152/12,000/1,040 simulation populations, and 70/70 G3-v2 certification.

Do not authorize GitHub publication, public supersession activation, or Power BI publication through this audit. If R26 passes, authorize only the next bounded integration step: regenerate the version-separated Power BI/repository candidate from exact R26 physical facts and submit that delta for appropriate review before publication.

ORIGINAL AUDIT DECISION / EXCERPTS

R26 focused successor audit completed

Decision ID: IA-D-R15-M12-001

Governing operating model: IA-GOV-20260822-001

Overall disposition: PASS_WITH_OBSERVATIONS

Acceptance includes observations. The review distinguishes the corrected policy, reproduced outputs, failed-run history and permission to integrate.

Governing decisions	
Matter	Independent disposition
Owner-approved R26 M2.5 monitoring policy	PASS
Physical R2 M2.5–M2.10 business outputs	PASS
R26 R2 runtime as a whole	FAIL — HISTORICAL PROVENANCE
Retained-final clone-and-resume architecture	PASS_WITH_OBSERVATIONS
R26 M2.11 eight-profile simulation	PASS
R26 M2.12 / G3-v2 certification	PASS
R26 technical successor baseline	PASS
Version-separated repository / Power BI integration	AUTHORIZED NEXT STEP

Preserve the failure. Prove the accepted result.

The corrected result passes review. An earlier failed run, identified as R2, stays in the record as a failure.

ORIGINAL AUDIT FINDING / EXCERPT

Why R26 passes

The policy implementation is exact. Independent static re-performance produced 16/16 PASS: monitoring starts from first expected remittance, expected pace is based on selected total repayment divided by selected horizon, WATCH eligibility begins at day 21, adverse pace eligibility at day 30, three-day persistence is enforced, and the existing early no-remittance, zero-sales, liquidity and horizon triggers remain present.

The physical 15,600-row R26 daily monitoring population also independently reproduces all 18 submitted monitoring checkpoints with zero mismatch. Raw overlapping alerts remain preserved while the operational alert-family layer separately consolidates them.

RETAINED FAILED RUN / SAME CAPTURE / EXCERPT

R2 failure classification

The R2 result correctly remains a failed runtime record:

R2 result SHA-256

f148d5f51d464b379dc218902f1b61844914a86207665b4ff0699aab8bbe4329

Status

FAIL

Failure stage

FINAL_M2_11_M2_12

That failure does not invalidate its completed Wave A business outputs.

R26 / AUDIT EVIDENCE

THE INTEGRITY TEST

Repeating the checks reproduces 16/16 policy checks and 18 monitoring checkpoints with no mismatch. The accepted business outputs agree; the earlier failure remains separately recorded.

PRACTICE TO REUSE

Accept the verified result without rewriting the history that produced it.

Verify the change. Reconcile what should stay the same.

Check the before-and-after monitoring results for the same 130 funded scenario paths. Confirm that cash history and end-of-period outcomes remain unchanged.

103 BASELINE FUNDED PATHS

41 → 20

BEFORE → AFTER

Baseline: peak adverse accounts

27 STRESS FUNDED PATHS

11 → 4

BEFORE → AFTER

Stress: peak adverse accounts

Adverse: underperforming, severe shortfall or dormant/no remittance; WATCH is excluded.

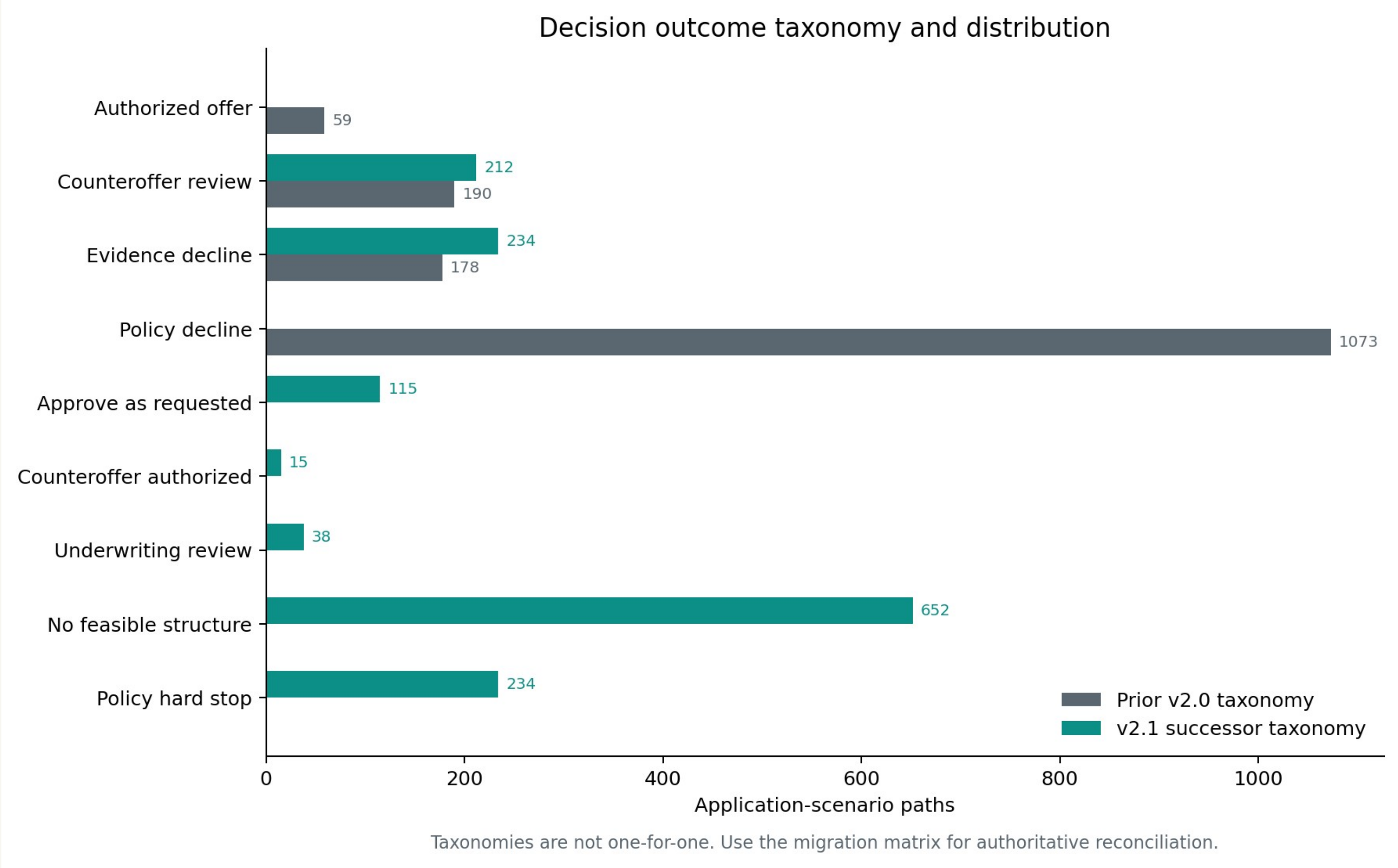
END-OF-PERIOD OUTCOME	VALUE - UNCHANGED BY CALIBRATION
Baseline paths paid off by period end	99 of 103 paths
Stress paths paid off by period end	27 of 27 paths
Baseline severe accounts at period end	4
Baseline attention exposure at period end	\$7,823.28

The policy changes when repayment pace is flagged; cash history and end-of-period results stay the same. This establishes better calibration; it does not establish improved collections.

Make the decision distinctions visible.

Return to the full decision-system comparison: the revised labels distinguish approvals, reviews, evidence gaps and policy stops.

ORIGINAL EXECUTIVE CHANGE BRIEF / FIGURE 3 / PDF EXCERPT



750 applications under two scenarios give 1,500 paths. Four broad categories become seven more specific outcomes. These are category totals; the migration matrix traces individual paths.

Show the search evidence behind the changed decision path.

The full rebuild examines more alternative financing structures and records why cases move into support or review.

557 → 6,519

reportable candidate structures

143

prior declines moved to support or review

ORIGINAL CHANGE LOG / SEARCH POPULATIONS

5.3 Candidate search and proof populations

Population	Prior v2.0.0	Successor	Interpretation
Application-scenario rows	1,500	1,500	Same comparison population
Candidate structures	557	6,519	Expanded governed search
Exact-search candidates	Not separately reported	6,397	Candidate alternatives generated by search
Requested references	Not separately reported	122	Requested-structure reference rows
Decision-gate rows	Not separately materialized	84,228	Constraint-by-candidate evidence
No-feasible proof rows	No separate proof population	652	Explicit failed-search evidence

Support or review is not authorization.

Product limits, economic hurdles and decision authority still apply. The 143 migrations include governed review; they are not 143 approvals.

Give another reader a way to challenge an outcome.

The later teaching artifact turns decision evidence into a repeatable explanation exercise.

ORIGINAL REPORT PAGE / COMPLETE PAGE

MSBF / MERCHANT FINANCE MASTERCLASS

TAKE THE REASONING WITH YOU

23

Understand the business.
Explain the decision.

Connect what is known, what is requested, what is permitted, and what happens next.

I READ THE MERCHANT

Sales start the question.

Usable evidence, cash flow and existing obligations shape capacity.

II READ THE REQUEST

Terms work together.

Evaluate amount, remittance, payback and horizon as one financing structure.

III SEARCH & DECIDE

Support is not authority.

A permissible alternative must still clear its final credit and judgment gates.

IV FOLLOW THE ACCOUNT

Funding starts a new question.

Monitor before acting. Reconcile events and balances. Explain what remains.

Do not stop at the outcome. Teach the reason behind it.

FINAL TRY IT

Choose one merchant case. Explain its outcome without saying “because the model said so.”

ANSWER / TEACH-BACK CHECK

Name the inputs, the binding rule, the permitted action and the account evidence.

Start with the merchant. Test the structure. Respect authority. Follow the account.

Andrew R. Goad · Synthetic MSBF · Non-production · No real funds

Teaching synthesis of lessons 01–22 | 23 / 23

Explain a case through its inputs, binding rule, permitted action and account evidence. The exercise makes the reasoning usable beyond the original conversations.

Use the same questions on the next analytical problem.

The case becomes a working discipline that another team can inspect and apply.

Frame What business behavior needs a better explanation?

Trace Which source information and rule produced the result?

Challenge Does the accepted rule make sense for the problem?

Test Which hypothesis separates the competing explanations?

Reconcile Do the full set of records and the facts that should stay unchanged still agree?

Review What can a separate reviewer verify, and what next step is permitted?

Return Does the report explain what the evidence actually supports?

Teach Can another person explain the result and its limits?

Stay accountable across the loop, including after an acceptance milestone.

The leadership method this case demonstrates.

Frame the decision, direct the work, challenge the evidence and make the reasoning usable.

Decision framing	Turn an unclear result into a specific question that the source evidence can answer. Evidence lineage and the challenge to accepted routing rules
AI-assisted delivery	Specify how work should proceed, be repeated and leave evidence for review. Cross-chat requirements, execution direction and no cold handoffs
Analytical challenge	Question persuasive answers and test competing explanations. Monitoring grain, physical QA and the timing hypothesis
Governed acceptance	Verify the result and distinguish technical acceptance from permission to proceed. R23 and R26 decisions, including retained failure provenance
Knowledge transfer	Make a complex system understandable enough to challenge and teach. Reporting requirements, corrected views and the teaching exercise

The system improved through a process that remained traceable, reviewable, teachable and bounded.

Do not stop at the output. Teach the reason behind it.

Ask what the system has established.

Check what remains unproven.

Keep judgment accountable to evidence.

Andrew R. Goad

Human-in-the-Loop Analytical Leadership

For leaders who need analytical systems they can challenge, explain and trust.

EXPLORE THE PROJECT AND PUBLISHED EVIDENCE

github.com/andrew-goad/merchant-sales-based-financing-strategy-simulator